

## Frozen MatterGen baseline

C0 · atomic species + positions + cell · CFG = 2.0

### INNOVATION 1 / ATTRIBUTE CONTROL

residual diagnostics + terminal selection

reference-preserved  
shared prefix

Fixed-K2 candidate branches

terminal constraints → exact C0 fallback

### INNOVATION 2 / LOCAL CONSISTENCY

stage reliability + force feedback

late-stage MatterSim  
force feedback

periodic mapping +  
bounded correction

position-only injection → safety fallback

## Unified frozen evaluation

paired seeds · MatterSim · CHGNet · surrogate boundaries